

Self as Integrated Whole: The Operational Specification of the Self as the Integrated CKS-Governed AI Intelligence That Holds Multiple Aspects as Facets Under Unified Governance

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), the second paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026). It does not introduce new axioms. Its sole contribution is to formalize, in operational form, the Paper 2 specification of the Self as integrated whole — the integrated CKS-governed AI intelligence that holds multiple aspects as facets under unified governance, where the instinct/reasoning separation per B1.01 operates as foundational architecture and all Paper 1 commitments hold at Self-distinct scope per B1.20. This is the twentieth note in Phase B2 of Series B and the first of five notes decomposing B1.05.

Abstract

Paper 2 introduces three architectural levels — cell, aspect, and Self — and specifies the Self as the integrated whole that holds multiple aspects as facets of one CKS-governed intelligence (B1.05). The Self is Paper 2's highest structural level and the level at which the instinct/reasoning separation per B1.01 operates as foundational architecture. Yet the Self is operationally distinct in a way that requires its own formalization. Conventional multi-component AI architectures use an "AI system" framing where the components together are treated as "the AI"; CKS instead specifies the Self architecturally — as an integrating frame with its own substrate, governance scope, and architectural identity, holding aspects as facets through which the integrated intelligence manifests rather than as subsystems composed into a sum. This note formalizes Self-as-integrated-whole as a standalone operational variant. It states the architectural specification, names what makes the Self architecturally distinctive, articulates the cognitive analog of integrated personhood as direct architectural realization, lists inherited Paper 1 commitments at Self level per B1.20, develops operational implications, gives an operational test, and opens the five-note B1.05 decomposition (B2.20 → B2.24).

1. Why Self-as-integrated-whole needs to be formalized as a standalone operational variant

Paper 2's three-level structure puts the Self at the top. The Self is the integrated CKS-governed AI intelligence holding multiple aspects as facets of one whole (B1.05). It inherits Paper 1's commitments through B1.20, runs the instinct/reasoning separation per B1.01 as foundational architecture, and operates over aspects as content domain through B1.18. None of these properties are added at the Self level — they are recursively inherited and configured. What is added at the Self level is the integrating frame itself: an architectural object that is more than a list of aspects but less than a deployment.

The Self is the most readily misread of Paper 2's three levels. Cells map cleanly to Paper 1; their architectural identity is established by A1.01–A1.16. Aspects are coordination arrangements with explicit purpose-defined membership; their character is operationalized by B2.15–B2.19. The Self has no direct Paper 1 antecedent and no obvious external referent — it is what *integrates* a CKS-governed AI intelligence into one whole, but is not identical to "the system," not identical to "the deployment," and not identical to any single aspect. Without standalone specification, downstream implementations are likely to drift toward one of three misreadings: (a) Self equals the sum of its aspects (collection-of-aspects misreading); (b) Self equals the deployment in which it runs (deployment-conflation misreading); or (c) Self is a top-level orchestrator that commands aspects (command-hierarchy misreading). Each misreading collapses an architectural commitment that B1.05 makes load-bearing. This note pins down the specification and opens the B1.05 decomposition that B2.21–B2.24 carry forward.

2. The architectural specification

The Self is the **integrated CKS-governed AI intelligence** as one architectural entity. It is the highest structural level in Paper 2, operationally specified by six properties.

Holds aspects as facets, not as subsystems. The Self holds multiple aspects (B1.04) as facets of one whole. The "facet" character is operationally distinct from "container" or "subsystem": aspects are not contents of the Self, and they are not separable subsystems composed into the Self. They are facets through which the integrated Self manifests for different purposes. The Self is not the sum of its aspects; it is the integrating frame within which aspects coexist.

Has Self-level substrate. The Self has its own substrate — substrate-resident authoritative content at Self scope per A2.46 (Category 4). Self-level substrate is not the union of aspect-level substrates and is not derived from cell-level content. It carries:

- *Self integration architecture* — how aspects are integrated, how Self-level operations span aspects, how aspect coexistence is governed (the subject of B2.21).
- *Instinct/reasoning configuration per B1.01* — which LLM instances function as instinct layer (per B2.01), which substrate constitutes reasoning layer (per B2.02). This is the subject of B2.23.
- *Self-level rules per A2.04* — orchestration rules at Self scope, distinct from aspect-level or cell-level rules.

- *Self-level outputs* — integrated outputs that represent the whole's response, often the deployment's externally visible behavior.
- *Self identity* — the architectural identity that makes a Self the same Self across aspect changes, cell reassignments, and vertical evolution per B1.16.

Operates over aspects as content domain per B1.18. The Self operates over aspects as content — asking pattern questions across aspects, selecting aspects for given purposes, governing coexistence — rather than commanding them.

Has cross-level access to cells per B1.19. The Self can access cells directly when purpose requires, without being routed through aspects. Cross-level access does not collapse the level distinction; it only confirms that the Self is not architecturally confined to operating only through aspects.

Preserves all Paper 1 commitments at Self-distinct scope per B1.20. The full A1.01–A1.16 commitment set holds at Self level (§5).

Has Self-distinct scope per B2.07. The Self level operates at the scope of integrated CKS-governed AI intelligence as one whole — distinct from cell-level scope and aspect-level scope, even though all three inherit the same Paper 1 commitments.

3. What makes Self-as-integrated-whole architecturally distinctive

Three properties distinguish Paper 2's Self from conventional multi-component AI framings.

The Self is a specified architectural entity, not an emergent system label. Conventional multi-component AI architectures often have an "AI system" framing — there are agents, models, tools, retrieval components, and orchestration layers, and "the AI" is the assembly. The assembly has no architectural specification of its own; it is what you get when you put the components together. CKS instead specifies the Self architecturally, with its own substrate, governance scope, and architectural identity. This is not a relabeling of the assembly; it is the specification of an integrating frame distinct from anything internal to it.

Aspects are facets, not subsystems. In subsystem framings, components are composed into a whole through interfaces, and the whole is the sum of the components. CKS aspects are facets through which the integrated Self manifests. The same underlying cells can participate in multiple aspects simultaneously through relational role membership; the Self does not own its aspects in the way a system owns its subsystems.

Integration is over coexistence, not over subordination. A subsystem-composed system integrates by hierarchy: the top-level component routes work to subsystems. The Self integrates by coexistence: aspects coexist as facets, Self-level substrate specifies how the coexistence is organized, and aspects are not subordinate to the Self in the way a subsystem is subordinate to its parent. Self-level rules govern aspect coexistence; they do not command aspect behavior. This is operationalized in B2.22.

4. The cognitive analog: integrated personhood with modes of engagement

The cognitive analog is direct. A person is one integrated whole, holding multiple modes of engagement — competitive-sports mode, calm-study mode, professional mode, parental mode — as facets of one personhood. The modes are not separate persons composed into the person; they are facets through which the integrated person engages different purposes. The person has identity as integrated personhood, not as the sum of modes. The same underlying capacities are recruited across modes, with the active mode determining how those capacities are expressed for the current purpose.

CKS Selves architecturalize this concept. The Self is the architectural realization of integrated personhood: the integrating frame within which aspects (modes of engagement, in the cognitive analog) coexist as facets. The same underlying cells can be recruited across aspects through relational role membership; the active aspect determines how those cells are recruited; the Self has its own identity as the integrated whole.

The analog functions as conceptual scaffold rather than as architectural derivation. The architectural substance is integrated CKS-governed AI intelligence at a level distinct from cells and aspects — substrate-resident, human-governed, conflict-preserving, AI-mediated, tool-agnostic, linear-cost — not the cognitive structure it parallels.

5. Inherited Paper 1 commitments at Self level

Per B1.20 (recursive Paper 1 commitments), each Series A commitment holds at Self-distinct scope per B2.07. Humans govern Self-level operations and Self-level rules at all times (A1.01); the substrate–entity boundary holds with Self-level substrate authoritative for Self-level content (A1.02); conflict at Self level is preserved as first-class substrate state, particularly across aspects (A1.03); AI mediates substrate at Self scope with the five mediator properties preserved (A1.04); Self-level substrate is tool-agnostic (A1.05); Self-level governance cost is not size-proportional — adding aspects or growing aspect-internal cell counts does not grow Self-level governance cost (A1.06); Self-level operations carry the six-field provenance metadata (A1.07); Self-level substrate is the source of truth for Self-level content (A1.08); the two-axis governance and multi-human structure holds at Self scope (A1.09); the five determinism guarantees apply to Self-level operations (A1.10); Self-level governance is realizable by non-specialists with appropriate access (A1.11); the three-mode labor allocation applies to Self-level labor (A1.12); the five composition constraints hold for Self-level composition (A1.13); the Self level is distinguishable from RAG, parametric memory, and external structured memory (A1.14); the Self level is distinguishable from workflow engines, agent frameworks, and control planes (A1.15); and the three composition patterns apply when the Self composes with adjacent AI components outside its boundary (A1.16). The recursive inheritance is what makes the Self level architecturally coherent with Paper 1; verification is the subject of B2.24.

6. Operational implications

Six implications follow from the specification.

Deployments configure the Self per requirements: what aspects the Self holds, what integration architecture organizes them, what instinct/reasoning configuration runs at Self level, what Self-level

rules apply, what Self-level outputs the Self produces. The architecture supplies the form; the deployment supplies the substance.

Selves are instantiated through governed birth per B1.09, applied at Self scope and operationalized in subsequent Phase B2 work on Self instantiation. Lifecycle operations (birth, mating, death) and vertical evolution (adding, removing, modifying aspects, modifying integration architecture) per B1.16 apply at Self level under the same authority architecture as cell and aspect lifecycle and evolution.

The Self may be the deployment-visible architectural entity: callers outside the Self interact with Self-level outputs, while aspects and cells operate within the Self under Self-level governance. This does not make Self equal to deployment — a deployment is a broader operational object that may include Self instantiation, infrastructure, hosting, and integration with non-CKS systems; the Self is the architectural integration of CKS-governed AI intelligence within that deployment.

When Selves span partner boundaries, A2.47 (cross-partner authority) governs Self-level authority across the boundary; cross-partner Selves remain integrated wholes, with the cross-partner concerns handled at the authority level. Self verification per B2.24 supplies the operational protocol for confirming inheritance under B1.20.

7. Limits

Eight limits bound the Self-as-integrated-whole specification.

The Self is not the deployment. A deployment is an operational object that hosts a Self; the Self is the architectural integration of CKS-governed AI intelligence within it.

The Self does not eliminate aspect-level or cell-level distinct scope. Per B2.07, each level has distinct scope; the Self level adds integrating-frame scope without absorbing the lower levels.

The Self does not command aspects. The Self–aspect relationship is the higher-as-content-domain relationship per B1.18 — operating over, not commanding. This is operationalized in B2.22.

The Self does not bypass Paper 1 commitments. All A1.01–A1.16 commitments hold at Self scope; the Self level introduces no architectural shortcuts that would collapse them.

The Self is not a monolithic intelligence. The integration is over coexisting aspects, each retaining distinct purpose, scope, and substrate content.

The Self does not prescribe specific aspect compositions. Aspect composition is a per-deployment design decision under orchestration substrate.

The Self is not a substitute for aspect-level or cell-level governance. Per A2.47, governance is distinct at each level and the three governance scopes coexist.

Paper 2 specifies INTRA-Self architecture only. Inter-Self dynamics — how multiple Selves coordinate, compete, cooperate, share substrate, or evolve in interaction — are out of scope for Paper 2 and addressed in Paper 3.

8. Operational test

A Self in a Paper 2 deployment satisfies the integrated-whole specification when all of the following hold simultaneously:

1. **Aspects are facets, not subsystems.** Multiple aspects coexist within the Self as facets through which the integrated intelligence manifests; the same underlying cells can participate in multiple aspects simultaneously through relational role membership.
2. **Self-level substrate exists with Self-distinct content.** The Self has substrate-resident authoritative content at Self scope per A2.46: integration architecture specifications, instinct/reasoning configuration per B1.01, Self-level rules per A2.04, Self-level outputs, and Self identity. This content is not the union of aspect-level substrates and is not derived from cell-level content.
3. **Self–aspect relationship is content-domain.** The Self operates over aspects as content domain per B1.18 — asking pattern questions, selecting aspects for given purposes, governing coexistence — and not as commanding parent.
4. **Cross-level access is preserved.** The Self can access cells directly per B1.19 when purpose requires.
5. **All Paper 1 commitments hold at Self-distinct scope per B1.20.** A1.01 through A1.16 are inheritable, audited, and active at Self level; B2.24 supplies the verification protocol.
6. **Instinct/reasoning separation per B1.01 operates at Self level.** The Self specifies which LLM instances function as instinct layer and which substrate constitutes reasoning layer; the separation is foundational architecture at the Self level, not added at aspect or cell level.
7. **Self has architectural identity distinct from any single aspect and from the deployment.** The Self is the integrating frame, with identity preserved across aspect changes and cell reassignments per B1.16. The Self is not coextensive with any one aspect, not coextensive with the sum of its aspects, and not coextensive with the deployment that hosts it.

A system that fails any of (1)–(7) has not realized Self-as-integrated-whole as Paper 2 specifies it, regardless of how well it satisfies any individual Paper 1 or Paper 2 commitment in isolation.

9. Why naming as standalone matters; opening the B1.05 decomposition

Naming Self-as-integrated-whole as a standalone operational variant has three consequences.

It pins the Self level against three predictable misreadings. The collection-of-aspects misreading, the deployment-conflation misreading, and the command-hierarchy misreading are the three ways the Self level is most readily collapsed. The standalone formalization names each, distinguishes the Self from each, and supplies the architectural substance the misreadings dilute.

It opens the five-note B1.05 decomposition. B2.20 (this note) formalizes the Self as integrated whole — the foundational operational specification. B2.21 will operationalize the Self integration architecture as substrate-resident specification. B2.22 will operationalize the Self–aspect content-domain relationship under B1.18. B2.23 will decompose the Self-level instinct/reasoning configuration per B1.01. B2.24 will formalize Self-level inheritance verification — the operational protocol for confirming all A1.01–A1.16 commitments hold at Self-distinct scope per B1.20. Subsequent Phase B2

notes (B2.25–B2.29) decompose B1.06 two-layers within every cell, and later notes proceed through the lifecycle and evolution machinery.

It supports the prior-art posture of the series. Each note formalizes one patentable derivation; the more notes, the smaller the territory where any party can claim novel invention without bumping into the prior-art chain. Self-as-integrated-whole is a derivation that downstream work in integrated AI architectures, multi-aspect AI Self construction, and AI personhood frameworks is likely to converge on; the standalone formalization places that convergence under prior art with the source paper's commitments.

10. Conclusion

Self-as-integrated-whole is the operational specification of the Self as Paper 2's highest structural level: the integrated CKS-governed AI intelligence that holds multiple aspects as facets of one whole under unified governance, with the instinct/reasoning separation per B1.01 operating as foundational architecture and all Paper 1 commitments holding at Self-distinct scope per B1.20. The Self has its own substrate (integration architecture, instinct/reasoning configuration, Self-level rules, Self-level outputs, Self identity), operates over aspects as content domain rather than as subordinates, and has architectural identity distinct from any single aspect and from the deployment that hosts it. The cognitive analog of integrated personhood with modes of engagement is a direct architectural realization, with the architectural substance being integrated AI intelligence at a level distinct from cells and aspects. The treatment opens the B1.05 decomposition that B2.21–B2.24 carry forward, and bounds Paper 2's scope to intra-Self architecture, leaving inter-Self dynamics to Paper 3.

Subsequent work that implements, extends, or argues against the Paper 2 Self should use *Self as integrated whole* in the sense formalized here. Subsequent work that uses the term differently is using a different concept, and the difference should be named.

Source paper

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance*. April 2026. ORCID: 0009-0004-8065-3235.

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